

Hema Praharsha Kanithi

Senior Cloud / Infrastructure Engineer

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PROFESSIONAL SUMMARY

Senior Cloud Engineer with 8+ years of experience designing, automating, and supporting enterprise infrastructure across AWS, Kubernetes, and hybrid environments. Skilled in building scalable, secure, highly available cloud solutions using Terraform, container orchestration (Docker, EKS), CI/CD automation, and cloud-native services to improve deployment efficiency and platform reliability.

Strong background in AWS infrastructure engineering, cloud migration, observability, incident management, and disaster recovery using EC2, VPC, IAM, Lambda, ECS, EKS, S3, CloudWatch, Route 53, Auto Scaling, and Systems Manager. Experienced collaborating with cross-functional engineering and security teams to modernize infrastructure and accelerate delivery through DevOps practices, with additional exposure to AWS AI/ML services supporting healthcare data solutions.

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate**
- **AWS Certified Developer – Associate**
- **Certified Terraform Associate**
- **AWS Certified Cloud Practitioner**

EDUCATION

Wright State University – Ohio, USA

Jan 2024 – May 2026

- Master's in Computer Science (MScS)

Andhra University – Andhra Pradesh, India

June 2015 – June 2018

- Bachelor's in Computer Science (B.Sc.)

KEY SKILLS

- **Cloud Platforms:** Amazon Web Services (AWS): EC2, VPC, IAM, S3, Route 53, Elastic Load Balancing (ELB), Auto Scaling, Lambda, ECS, EKS, RDS, DynamoDB, CloudWatch, Systems Manager (SSM), Secrets Manager, SNS, SQS, API Gateway, CloudFront, AWS DataSync, AWS Glue, Amazon MWAA (Apache Airflow), FSx, EventBridge
- **Infrastructure as Code & Automation:** Terraform, Terraform Modules, Terraform Workspaces, Ansible, Infrastructure as Code (IaC), Policy as Code, Terraform Sentinel
- **Containers & Platform Engineering:** Docker, Kubernetes, Amazon EKS, OpenShift, Helm, Container Orchestration, Cluster Administration, Horizontal Pod Autoscaling, Cluster Autoscaler, Service Discovery, Ingress Management, Platform Operations
- **CI/CD & DevOps:** Jenkins, GitHub Actions, GitLab CI/CD, Release Automation, Deployment Pipelines, Blue-Green Deployments, Rolling Deployments, DevSecOps
- **Monitoring & Observability:** Amazon CloudWatch, Prometheus, Grafana, Splunk, New Relic, LogDNA, Log Aggregation, Application Monitoring, Infrastructure Monitoring, Dashboards, Alerting, Incident Response, Root Cause Analysis

- **Security & Governance:** AWS IAM, HashiCorp Vault, AWS Secrets Manager, Role-Based Access Control (RBAC), Least Privilege Access, Security Policies, Compliance Automation, Encryption at Rest & In Transit
- **Networking & Cloud Architecture:** VPC Design, Subnets, Route Tables, NAT Gateway, VPN Connectivity, Security Groups, Network ACLs, DNS Management, Hybrid Cloud Connectivity, Multi-AZ & Multi-Region Deployments, High Availability, Disaster Recovery
- **Programming & Scripting:** Python, Shell Scripting, Bash, REST APIs, Event-Driven Automation, YAML, JSON
- **Databases & Storage:** Amazon RDS, DynamoDB, PostgreSQL, Db2, Amazon S3, Data Migration, Data Validation, Backup & Recovery
- **AI/ML & Analytics:** Amazon SageMaker, AWS Glue, Amazon QuickSight, Machine Learning Pipelines, Model Training, Model Evaluation, Feature Engineering, Data Validation, Healthcare Analytics
- **Tools & Methodologies:** Git, Jira, ServiceNow, Linux, Apache HTTP Server, Apache Tomcat, Nginx

WORK EXPERIENCE

Senior Cloud Engineer – CVS Health

Phoenix, Arizona

Jan 2025 – Apr 2026

- Developed reusable Terraform modules and IaC standards to provision Amazon VPCs, subnets, security groups, IAM roles, Auto Scaling Groups, ELBs, and EKS infrastructure, managing remote state, state locking, and environment isolation across multiple AWS accounts.
- Designed and administered containerized workloads on Amazon EKS using Docker and Helm, delivering resilient, production-ready application deployments across development, QA, and production environments.
- Managed multi-tenant Amazon EKS clusters supporting 40+ microservices across 15+ namespaces, implementing Horizontal Pod Autoscaling and Cluster Autoscaler policies that improved compute resource utilization by approximately 25%.
- Standardized reusable Helm chart templates and Kubernetes RBAC policies across 20+ service accounts, cutting new service onboarding time from days to under 2 hours while enforcing least-privilege access on EKS.
- Implemented Kubernetes-native rolling and blue-green deployment strategies via GitHub Actions and Jenkins, sustaining 99.9% uptime across production EKS workloads and cutting failed deployment rollbacks by approximately 40%.
- Architected highly available AWS infrastructure using EC2, Auto Scaling, ELB, Multi-AZ deployments, Route 53, and NAT Gateway to support CVS Caremark PBM platforms with fault tolerance and business continuity.
- Automated infrastructure provisioning and deployment workflows using GitHub Actions, Jenkins, and GitLab CI/CD, enabling IaC validation and consistent releases across AWS environments.
- Built enterprise observability using CloudWatch, Prometheus, Grafana, and Splunk, developing dashboards and alerting strategies that improved platform visibility and reduced incident response time by approximately 30%.
- Strengthened security and governance using AWS IAM, HashiCorp Vault, Secrets Manager, and Terraform Sentinel, enforcing least-privilege access and HIPAA-aligned compliance controls across production infrastructure.

Cloud AI/ML Engineer Intern – Dynamic Tech Work

San Francisco, CA

May 2024 – Aug 2024

- Built an end-to-end AWS-native predictive healthcare analytics solution for 30-day patient readmission risk using S3, AWS Glue, SageMaker, Lambda, QuickSight, IAM, and CloudWatch, adhering to HIPAA data handling standards.
- Engineered ETL pipelines in AWS Glue and performed feature engineering on patient demographics, diagnoses, and lab results to improve model accuracy and data quality.
- Built and evaluated ML models in SageMaker (Random Forest, XGBoost, Logistic Regression), comparing performance via Accuracy, Precision, Recall, F1-Score, and ROC-AUC to select the best model.
- Automated serverless ML workflows using Lambda and CloudWatch, and developed QuickSight dashboards visualizing patient risk stratification for healthcare stakeholders.

AWS Cloud Engineer – Accenture Solutions Pvt. Ltd

Hyderabad, India

Oct 2022 – Jan 2024

- Architected an AWS cloud migration platform for Veeva Vault, enabling secure migration of 50+ TB of multi-terabyte life sciences data across 20+ workloads using EC2, ECS, Docker, ECR, Lambda, S3, MWAA (Apache Airflow), DataSync, RDS, and Terraform.
- Engineered reusable Terraform modules to provision VPCs, subnets, security groups, IAM roles, ECS services, and MWAA environments, standardizing deployments across dev, test, and production.
- Designed event-driven serverless architectures using Lambda, CloudWatch, SNS, DynamoDB, and Systems Manager Parameter Store to automate workflow orchestration and centralized configuration.
- Implemented secure hybrid-cloud migration using AWS DataSync, Transfer Family, S3, and VPN connectivity to transfer and reconcile multi-terabyte datasets from on-premises to AWS.
- Automated data validation and reconciliation using Lambda, RDS, and DynamoDB, ensuring migration accuracy and regulatory compliance for business-critical life sciences data.
- Automated CI/CD deployments using Terraform, Docker, ECR, and ECS across environments, performing release coordination, rollback support, and production health checks.

Software Engineer II – Amazon India Pvt. Ltd.

Hyderabad, India

Sept 2021 – May 2022

- Supported mission-critical Amazon Go store platforms by monitoring production systems, troubleshooting software and device incidents, and restoring service availability across distributed retail environments.
- Investigated production issues through log analysis and root cause analysis, partnering with engineering teams to validate defects and drive permanent resolutions.
- Enhanced SOPs, operational runbooks, and troubleshooting documentation, improving support efficiency and reducing incident resolution time.
- Delivered Knowledge Transfer sessions and technical guidance to team members on production workflows and troubleshooting methodologies.

Technical Support Analyst – SourceOne Technologies (Deloitte)

Hyderabad, India

Dec 2020 – Sept 2021

- Supported production applications for The Hartford Insurance as SME, monitoring system health and performing root cause analysis to restore application availability.
- Coordinated production deployments across Development, UAT, Pre-Production, and Production using Jenkins, executing release verification and post-deployment health checks.

- Supported AWS cloud operations including EC2, S3, IAM, and CloudWatch, monitoring infrastructure health and managing access controls.
- Participated in Disaster Recovery planning and failover validation, and automated routine operational tasks using Shell scripting to reduce manual effort.
- Managed incidents and change requests using ServiceNow, Jira, New Relic, and Splunk, developing SOPs and runbooks that improved support consistency.

Application Support / DevOps Engineer – IBM India Pvt. Ltd.

Hyderabad, India

May 2019 – Aug 2020

- Supported enterprise production applications through incident troubleshooting, root cause analysis, and Db2 data validation, ensuring high availability across business-critical environments.
- Coordinated application deployments across Development, UAT, Pre-Production, and Production using Jenkins, executing release verification, rollback support, and post-deployment health checks.
- Contributed to migrating applications from IBM WebSphere and Db2 on-premises infrastructure to Red Hat OpenShift, supporting hybrid cloud modernization.
- Participated in Severity-1 incident response and Disaster Recovery drills with Incident Management teams, minimizing business disruption.
- Used Jenkins, Ansible, New Relic, LogDNA, ServiceNow, and Jira for operational monitoring and release management; recognized with the Performance Champion Award for migrating 15+ applications to Red Hat OpenShift with zero unplanned downtime.

ML Data Associate – Amazon India Pvt. Ltd.

Hyderabad, India

Aug 2018 – May 2019

- Processed, annotated, and validated large-scale audio datasets for Amazon Alexa voice services, performing wake-word detection and speech transcription to improve model accuracy.
- Executed quality assurance on annotated datasets, identifying inconsistencies and ensuring compliance with annotation guidelines for AI/ML model training.
- Collaborated with ML and engineering teams to review annotation quality and improve data preparation workflows supporting NLP and voice recognition systems.

System Administration Intern – Reinhart Global Solutions

Hyderabad, India

May 2017 – Mar 2018

- Assisted in Linux server administration including system configuration, user/permission management, and routine health monitoring to ensure server availability.
- Configured core infrastructure services including DNS, NFS, Apache HTTP Server, Apache Tomcat, and Nginx, supporting application deployments and production maintenance.
- Developed Shell scripts to automate routine administrative tasks and supported firewall configuration and system patching for infrastructure security.